

Memo

To: EDRM 718 Analysts

From: Mike Seaman

Date: 24 April 2025

Re: Job Preparation Project Analysis

We have been contracted by RWS, a leading research firm, to assist with data analysis on a project that investigates the perceptions of graduating seniors regarding their prospects to be employed in a position that matches their educational preparation and hopes about a first job. Please see the accompanying letter from the client for more information. This memo is to highlight some of my own expectations about this project.

- Create a private repository for this project and invite me as a collaborator. Use my FourthZ GitHub username for the invitation. I will not alter your repo, other than to provide my comments via a pull request when the project is complete. Your invitation will allow me to clone your repository. Please make it easy for me to locate materials by keeping the repo neat, organizing your work into subfolders, and using file names that help me identify what is in each file.
- It is my understanding that all data were keyed in by hand, so obviously we will need to carefully check both primary and foreign keys.
- I do not think it is a good idea to average first and second semester GPAs unless we verify consistency. Please visually examine consistency. Also, let's exclude those students who are among those with the ten percent largest residuals when predicting second semester GPA from first semester GPA so that we can be even more comfortable about averaging the two semesters. Highlight this in a graph so that I can better explain the issue to the client.
- When you average GPAs, please use a weighted average based on the number of credits taken in a semester. This makes more sense than treating the GPAs equally.
- When comparing job preparation perception across discipline categories, first test the omnibus hypothesis about mean equality, but then construct confidence intervals for all pairwise differences. Please keep our total confidence at 95% for all inferences.
- There are at least two functions that could be useful not only for this project, but for other contracted projects we may receive in the future. I have described these in a separate document. I have already written my versions of these functions, so if you decide not to write your own functions you may email me to obtain my functions. Source these, and other functions you decide to write, into your main analytic notebook. This will help keep the notebooks more readable when I review them.
- Your main notebook does not need to be a formal report. I will glean from the work of all associates to create the final report, so you can write the notebook for me. You will still need to keep the notebook neat and organized. Provide comments for each graph, table, and analysis to assist me with understanding these outcomes.

I look forward to seeing your work. I'm glad that you are on my team!